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Bird Species Detection using Deep Learning

Megh Patel [AU2544020] & **Konark Karia** [AU2544011]

School of Engineering and Applied Science, Ahmedabad University

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Outline

Introduction

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- The project focuses on the domain of **computer vision** and **deep learning**, specifically applied to **bird species detection and classification.**
- The goal is to automatically identify bird species from images using advanced neural network models such as the **Vision Transformer (ViT) Models**.
- This problem is significant for applications in **biodiversity monitoring**, **wildlife conservation**, and **ecological research**, where manual identification is time-consuming and prone to error.

Introduction

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- High-resolution bird images are analyzed with advanced neural networks, including **Vision Transformers (ViT) architectures**.
- The system extracts discriminative features to classify species accurately and efficiently.

Motivation

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- Identifying bird species manually is challenging and requires expert knowledge, making it difficult for large-scale biodiversity monitoring.
- Large-scale biodiversity monitoring benefits from automated recognition.
- The growing availability of bird image datasets inspired the idea of applying deep learning to automate species recognition.
- A personal interest in artificial intelligence and its applications in environmental conservation motivated the development of this project.

Motivation

- After reviewing several research papers, we wanted to find a better and more efficient solution for bird species detection.

Literature Review

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Bird Species Categorization Using Pose-Normalized Deep Convolutional Nets [Branson et al.] [2014]:

- Introduces pose-normalized CNN [Convolutional Neural Network] where bird images are aligned using part features before extraction.
- Reduces viewpoint variation $\rightarrow$ better fine-grained discrimination.
- Achieves **75.7% top-1 accuracy** on CUB-200-2011.

Literature Review

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Deep Convolutional Networks for Fine-Grained Recognition (VGG Baselines) [Simonyan & Zisserman] [2014]:

- VGG-16/19 [Visual Geometry Group] serve as strong feature extractors for FGVC [Fine-Grained Visual Classification] tasks.
- Depth + small filters improve hierarchical feature learning.
- Achieves:
  - **70%** (VGG-16)
  - **72%** (VGG-19)

Literature Review

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Look Closer to See Better: Recurrent Attention for Fine-Grained Categorization [Fu et al.] [2017]:

- Proposes RA-CNN [Recurrent Attention Convolutional Neural Network] with multi-stage attention zooming into informative regions.
- Learns discriminative regions without part annotations.
- Reports **85.3% accuracy** on CUB-200-2011.

Literature Review

Learning Multi-Attention CNN (MA-CNN) for Fine-Grained Recognition [Zheng et al.] [2017]:

- Learns multiple attention regions automatically without part labels.
- Jointly learns part localization + classification.
- Achieves **86.5% accuracy** on CUB-200-2011.

Literature Review

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TransFGVC: Transformer-Based Fine-Grained Visual Classification [Shen et al.] [2022]:

- Uses Swin Transformer + LSTM [Long Short-Term Memory] refinement for contextual token learning.
- Suppresses background noise and captures fine details.
- Achieves **92.58% accuracy** on CUB-200-2011.

Literature Review

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SupCon-ViT: Supervised Contrastive Learning for Ultra FGVC [Lu et al.] [2023]:

- Supervised contrastive loss clusters same-class embeddings tightly.
- Effective for ultra-fine-grained species with minimal inter-class differences.
- Outperforms baseline ViT models significantly.

Literature Review

Dataset Flaws, Annotation Errors, and Mislabeled Images [Van Horn et al.] [2015]:

- CUB-200-2011 has **4% mislabeled images** ($\approx$ 471).
- ImageNet FGVC classes also show $\geq$ 4% label noise.
- Flickr dataset retrieval is only **84% correct** ($\Rightarrow$ 16% mislabeled).
- Label noise limits maximum achievable FGVC accuracy.

Dataset Details

- The project uses the **CUB-200-2011 (Caltech-UCSD Birds)** dataset, a standard benchmark for fine-grained bird species classification.
- It contains **11,788 images** across **200 bird species**, each annotated with **bounding boxes**, **part locations**, and **class labels**.
- The dataset is divided into **5,994 training images** and **5,794 testing images**, ensuring a balanced split for effective model training and evaluation.
- The images are diverse in pose, background, and lighting, providing realistic challenges for model generalization.

Methodology

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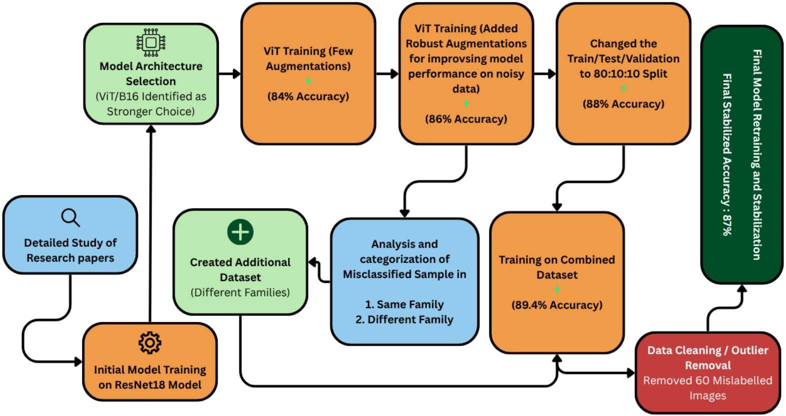

Figure 1: Workflow Flowchart

Methodology Overview

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- Comprehensive workflow covering:
  1. Dataset Preparation
  2. Data Cleaning (Mislabelled Image Removal)
  3. Manual Dataset Extension
  4. Data Preprocessing & Augmentation
  5. Model Selection
  6. Training Strategy
  7. Evaluation & Metrics

Dataset Preparation

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1. **Primary Dataset:** CUB-200-2011
   - Includes 11,788 images across 200 bird species.
   - Metadata files merged into a unified DataFrame:
     - Image paths
     - Class labels
     - Train/test split flags
2. **Shuffle and Stratified Splitting**
   - Split into:
     - 80% Training
     - 10% Validation
     - 10% Testing
   - Stratified by class to maintain distribution.

Dataset Preparation

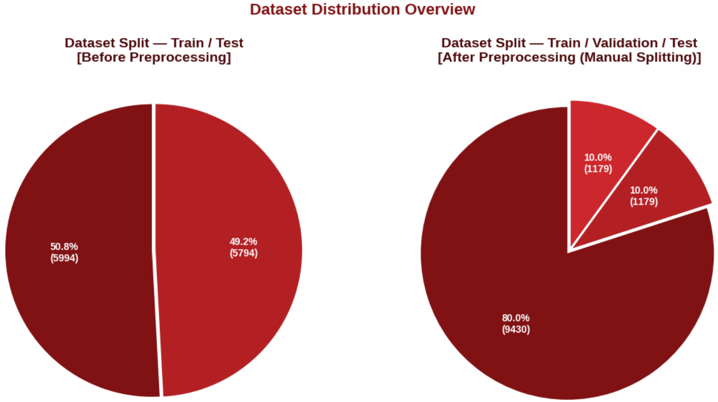

Figure 2: Dataset Split Distribution – Before & After Preprocessing

Data Cleaning

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- A curated list of **60+ mislabelled or out-of-distribution images** was identified.
- Prior research reports a **4–5% label error rate**, which aligns with our observations.
- All identified images were removed automatically during preprocessing.
- A detailed log file was saved for audit and reproducibility.
- Ensures a cleaner dataset $\rightarrow$ **improved training stability and accuracy**.

Dataset Extension: Manual Image Additions

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- Additional manually curated images were added to improve class balance and representation.
- We manually created the metadata files for these images and then imported them into the dataset pipeline.
- Added exclusively to the training set for enhanced model robustness.

Data Preprocessing & Augmentation

- **Purpose:**
  - Improve generalization
  - Prevent overfitting
  - Match ViT-B/16 input specifications

- **Transformations used (Training):**
  - Resize to 256
  - RandomResizedCrop(224, scale = 0.8–1.0)
  - Random Horizontal Flip
  - Random Vertical Flip (p = 0.1)
  - Random Rotation (±20°)
  - Color Jitter (brightness, contrast, saturation, hue)
  - Normalize (ImageNet mean and std)

Data Preprocessing & Augmentation

- **Validation/Test Transformations:**
  - Resize $\rightarrow$ CenterCrop(224)
  - Normalize

Data Preprocessing & Augmentation

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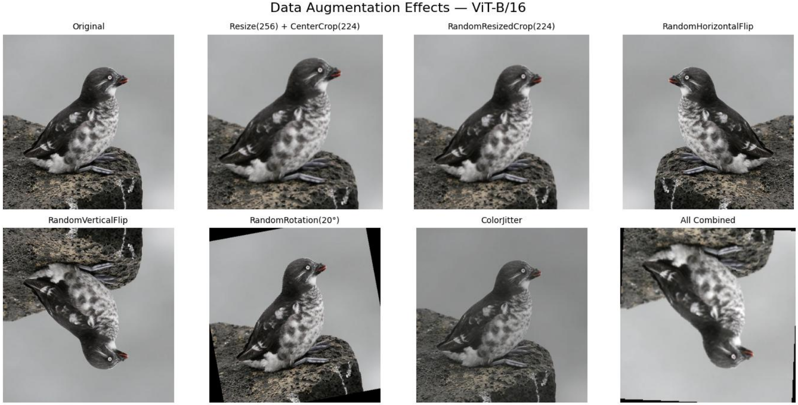

Figure 3: Data Augmentation Effects

Model Selection

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- **Model Used:** ViT-B/16 (Vision Transformer)
- **From:** TIMM Library
- **Training Procedure:**
  - Start with pretrained ImageNet weights
  - Freeze entire backbone initially
  - Train only the classification head (head LR = 3e-4)
  - Unfreeze **after epoch 5** for full fine-tuning
  - Continue fine-tuning with a lower LR (1e-5)
- **Loss:** CrossEntropyLoss
- **Optimizer:** AdamW (weight_decay = 0.01)
- **Scheduler:** CosineAnnealingLR

Training Strategy

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- **Mixed Precision Training:**
  - Reduces GPU memory usage
  - Speeds up training
- **Training Loop:**
  1. Forward pass
  2. Loss calculation
  3. Backward pass (scaled gradients)
  4. Gradient clipping (max norm = 1.0)
  5. Optimizer step

Training Strategy

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- **Validation per epoch:** Track accuracy & loss
- **Early Stopping:**
  - Patience = 6 epochs
  - Triggered if validation loss does not improve
- **Checkpointing:**
  - Best model saved based on min validation loss

Testing & Evaluation

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1. Load best-performing checkpoint
2. Run inference on test set
3. Compute:
   - Test Loss
   - Classification Report
   - Confusion Matrix
4. Save predictions and labels for further analysis

Progress Till Now

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- Conducted a detailed study of research papers that used the **CUB-200-2011 dataset** for bird species detection which achieved high accuracy.
- Gained a deeper understanding of the dataset through its metadata and supporting documentation.
- Initially trained the model using the **ResNet18 architecture**, but it achieved only **55% accuracy**.
- Later identified that the **ViT/B-16 model (pretrained on ImageNet)** would be a stronger choice for this task.

Progress Till Now

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- With minor **augmentations (Resize, Horizontal Flip)**, the ViT model achieved **84% accuracy**, motivating further exploration in this direction.
- Added additional **augmentations (Resize, ResizedCrop, Horizontal Flip, Vertical Flip, Rotation, ColorJitter)** to make the model robust to noise and orientation variations, improving accuracy to **86%.**
- Observed that the dataset contained comparatively fewer training samples, so we implemented a custom **80:10:10 training–validation–testing split**, reaching **88% accuracy**.

Progress Till Now

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- Analyzed and categorized the misclassified samples into:
  - **Birds belonging to the same family** (visually very similar)
  - **Birds belonging to different families** (identified using tokenized image names)
- Since intra-family species had minute visual differences, the model struggled with them. This led us to **create an additional dataset** focusing on **misclassified birds from different families.**
- Training the model on the **combined dataset (original + additional images)** increased accuracy to **89.4%.**

Progress Till Now

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- Further analysis revealed that some images in the main dataset were **mislabeled.**
- Researching other papers confirmed that the dataset had a **4–5% labeling error rate.**
- Through a detailed EDA process, we identified **60 mislabeled images**, which we completely removed from training, validation, and testing. After removing these, **the final model accuracy stabilized at 87%** within **5 epochs**.

Results Overview

- Achieved a final **87.12% test accuracy** on the CUB-200-2011 dataset.
- Training showed smooth **loss convergence** and improving accuracy across epochs.
- Significant improvement after **preprocessing and augmentation**.
- Error analysis reveals **70.2% of misclassifications occur within the same family**.
- Detailed visual results such as loss curves, accuracy trends, and confusion matrix are presented in the following slides.

Loss Graph

Figure 4: Training Validation and Test Loss Graph

Accuracy vs Epoch Graph

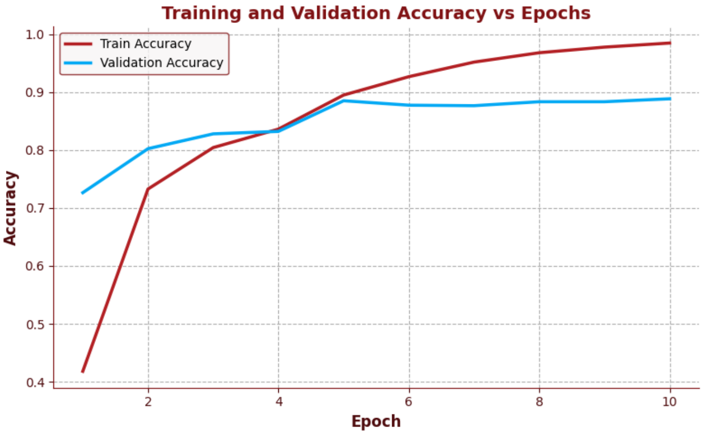

Figure 5: Training & Validation Accuracy vs Epoch

Confusion Matrix

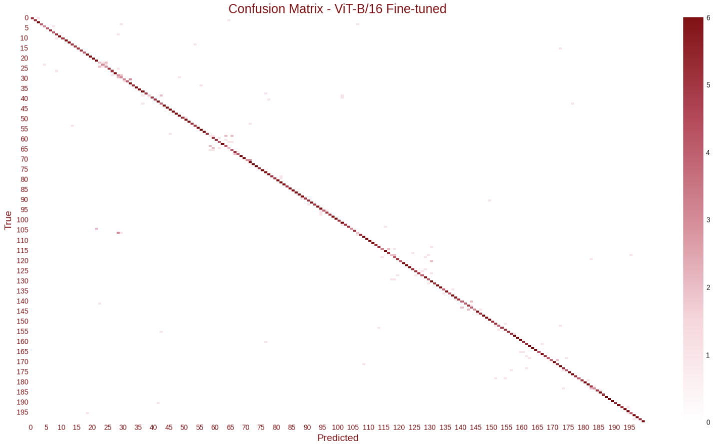

Figure 6: Class-Based Confusion Matrix

Misclassification Analysis — Family-level

- Extracted “family tokens” from species names.
- Compared:
  - True family vs predicted family
- Computed:
  - **70.2 %** of misclassifications within same family
  - **29.8 %** across different families

Misclassification Analysis — Family-level

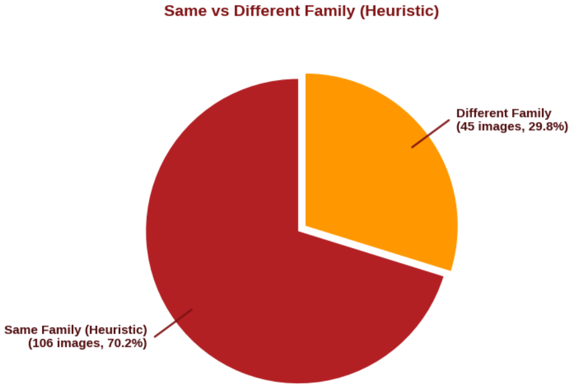

Figure 7: Same vs Different Family Distribution for Misclassified Images

Misclassified Samples

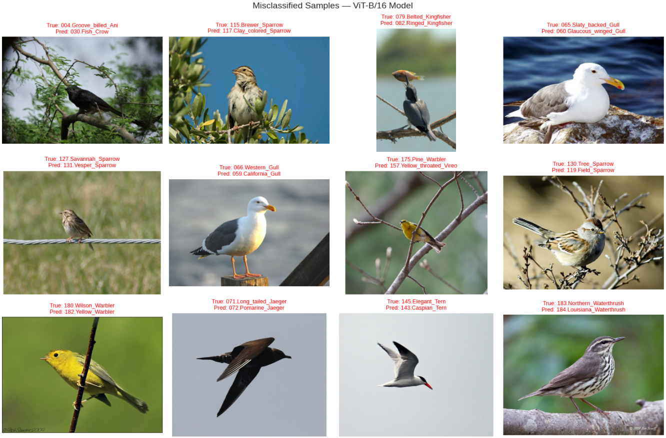

Figure 8: Misclassified Samples

Web UI

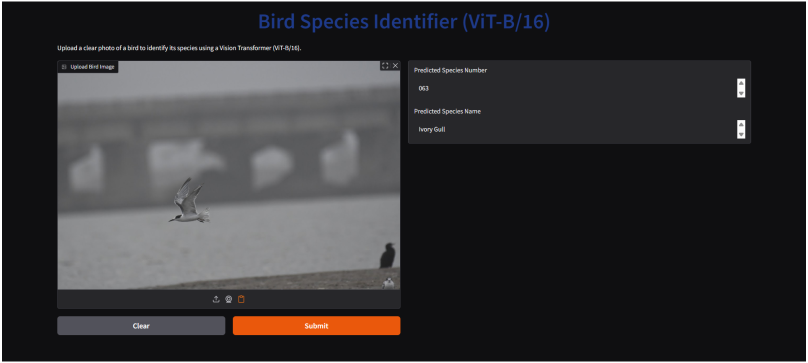

Figure 9: Web UI

Expected Future Work

- **High-Quality Manually Curated Dataset**
  - Manually inspect all images and remove mislabeled or out-of-distribution samples.
  - Add verified images from reliable sources (e.g., iNaturalist, GBIF).
  - Maintain metadata documenting inclusion/exclusion decisions.
  - Ensures a cleaner dataset leading to more stable training and higher accuracy.

Expected Future Work

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- **Improve Generalization on Additional Images**
  - Test model on external real-world bird images.
  - Add advanced augmentations (blur, lighting shift, background variation).
  - Use Test-Time Augmentation (TTA) for robust predictions.
  - Reduces overfitting and improves real-world performance.

Expected Future Work

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- **Better Classification Within Same Bird Family**
  - Use attention-based models to capture subtle visual differences.
  - Part-based learning: focus on head, wings, tail, feet regions.
  - Apply hierarchical classification (Family $\rightarrow$ Species).
  - Use feature-enhancement methods like Grad-CAM-guided fine-tuning.

References I

Papers

- Branson et al. (2014). *Bird Species Categorization Using Pose-Normalized Deep Convolutional Nets*
- Fu et al. (2017). *Look Closer to See Better: Recurrent Attention for Fine-Grained Categorization*.
- Simonyan & Zisserman (2014). *Deep Convolutional Networks for Fine-Grained Recognition (VGG Baselines)*.
- Zheng et al. (2017). *Learning Multi-Attention CNN (MA-CNN) for Fine-Grained Recognition*.
- Lu et al. (2023). *SupCon-ViT: Supervised Contrastive Learning for Ultra FGVC*.

Thank You!

Questions and Discussions

Presented by

Megh Patel & Konark Karia

School of Engineering and Applied Science, Ahmedabad University